ADVENTUREWORKS REPORT (HACKATHON SUBMISSION)

A BUSINESS PERFORMANCE STORY

SUBMITTED BY

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1.0 SUMMARY

Using a line-level extract from AdventureWorks2022, I built an Excel-based analytics pipeline (SQL → Power Query → Power Pivot → PivotTables → Dashboard) and produced a concise business dashboard showing revenue, profit, discount impact, territory performance, customer segmentation, and seasonality. From the analysis I derived five high-impact insights that identify profit leaks, channel opportunities, and operational strengths. Recommended actions focus on

- Optimizing discount policies
- Rebalancing reseller incentives
- Shipping cost optimization for specific territories
- Prioritizing high-margin product promotion,
- Seasonal inventory and marketing planning.

2.0 DATA & SQL (SOURCE USED)

2.1 Data Extraction (SQL)

I used the **AdventureWorks2022** database. I wrote a SQL query that joined multiple tables including:

- Sales
- Product
- Customer
- Person
- Store
- Territory

The query collected detailed product information, customer attributes, sales order header details, line-level details, and territory information.

I applied one filter:

WHERE p.StandardCost > 0

After running the script in SSMS, I **copied the results directly into Excel**.

2.2 Excel Workbook Structure

After loading the SQL output into Excel, I structured my workbook using four key sheets:

I. **Raw Working File** – SQL data pasted directly.

II. **Working Sheet** – cleaning, transformations, and new columns.

III. **PivotTables** – all PivotTables stored and clearly labeled.

IV. **Dashboard** – final visual dashboard built from PivotTables and measures.

This structure ensured my work remained clean, organized, and easy to maintain.

2.3 Data Transformation

1. **Power Query**: imported the SQL extract, converted date types, ensured numeric columns are numeric, trimmed strings, removed nulls where necessary.
2. **Feature engineering (Power Query / Excel calculated columns)**:
    - ProfitPerUnit = UnitPrice - StandardCost
    - COGS = StandardCost * OrderQty
    - DiscountAmount = (UnitPrice * UnitPriceDiscount) * OrderQty
    - TotalProfit = LineTotal - COGS
    - RevenueWithoutDiscounts = UnitPrice * OrderQty
    - RevenuePotentialLoss = RevenueWithoutDiscounts - LineTotal
    - DeliveryDays = ShipDate - OrderDate
    - ShippingStatus = IF(ShipDate <= DueDate, "On Time", "Late")
3. **DAX measures**: TotalRevenue, TotalProfit, TotalDiscount, AvgDeliveryDays, MarginPct, RevenueLostToDiscounts.
4. **PivotTables & Charts**: created separate sheets (Working Sheet, Pivot Tables, Dashboard) and linked PivotCharts, added slicers (OrderYear, ProductCategory), and conditional formatting for heatmap effect in the pivot.

3.0 BUSINESS INSIGHTS

3.1 **Insight 1 - Discounts are eroding profit without increasing volume**

Finding:
Aggregate measures on the dashboard show a substantial RevenuePotentialLoss and TotalDiscount while OrderQty is not higher for discounted lines, indicating discounts are not producing incremental volume but are reducing margin.

Evidence to show (dashboard):

- KPI on left: Potential Loss and Total Profit cards from summary pivot table and representative pivot table.

SQL/calculation used:

- PotentialLoss = SUM(UnitPrice * OrderQty) - SUM(LineTotal)
- Discount check: AVG(CASE WHEN UnitPriceDiscount > 0 THEN OrderQty ELSE NULL END) - compare with non-discounted average order quantity.

Interpretation:
Blanket discounts reduce gross profit and do not lead to higher units sold, company is sacrificing margin for no volume benefit.

Recommendation (priority):

1. Stop blanket discounts; introduce targeted, conditional discounts (e.g., min-order value).
2. Run randomized A/B tests on discounting to measure true incremental sales.
3. Reallocate discount budget to marketing/SEO or product bundling with higher ROI.

3.2 **Insight 2 - Individual customers are more profitable than stores (resellers)**

Finding:
Individuals generate higher revenue and profit per customer than Store customers, who often receive larger discounts which erode reseller profitability.

Evidence to show (dashboard):

- Bar Chart (Profit by Customer type) – Individuals customers give more profits that stores.

Calculation used:

- PivotTable - Profit by Customer Type

- **Interpretation:**
  Reseller discounts do not produce expected bulk volume; individuals are the real revenue drivers.

Recommendation (priority):

1. Reassess reseller discount policy; introduce performance-based tiers (discount only when minimum volume thresholds are met).
2. Invest in D2C retention strategies (loyalty programmes, personalized recommendations).
3. For underperforming resellers, apply stricter contract terms or channel re-evaluation.

3.3 **Insight 3 - Mountain Bikes deliver higher profit despite Road Bikes having higher revenue**

Finding:
Although Road Bikes show higher total revenue, Mountain Bikes deliver better aggregate profit (higher margin).

Evidence to show (dashboard):

- Bar chart(Top profitable subcategories): Revenue vs Profit by ProductSubcategory (Top left).

Calculation used:

- Pivot Table - Top 10 profitable Subcategories

Interpretation:
Road Bikes may be sold with greater discounting or higher COGS; Mountain Bikes are better margin performers.

Recommendation (priority):

1. Run price elasticity tests on Road Bikes to reduce discounting or reprice where possible.
2. Increase marketing investment in Mountain Bikes to scale profitable volume.
3. Optimize cost for Road Bikes (supplier renegotiation, bundling).

3.4 **Insight 4 - Territory effect , Australia outperforms due to lower freight**

Finding:
Australia shows highest revenue and better margin outcomes; freight cost is lower there compared to other territories, resulting in more competitive final prices and higher demand.

Evidence to show (dashboard):

- Chart: “Effect of Freights on Revenue” (center line chart showing total orders, profit and average freight).

SQL/calculation used:

- Pivot Table: Effect of freight

Interpretation:
Freight is a significant cost component; territories with lower freight show naturally higher sales.

Recommendation (priority):

1. Negotiate better freight terms or introduce local distribution in high-cost areas.
2. Introduce regional shipping surcharges or bin-based shipping strategies to reflect real cost-to-serve.
3. Test free-shipping thresholds strategically (e.g., above $X order value).

3.5 **Insight 5 - Strong seasonal pattern, Q1 is the major buying window**

Finding:
Bicycles, accessories (helmets, shorts, vests) and other related items spike in Q1 and decline through Q2–Q4. Shorts and vests have particular Q1–Q2 strengths.

Evidence to show (dashboard):

- Quarterly heatmap on the pivot table (right) and line chart of Profit per Month (top left).
- Category-specific pivot for Clothing & Accessories showing quarters.

Calculation used:

- Pivot Table: Total revenue per Quarter

  Interpretation:
  Customers make major lifestyle purchases early in the year (New Year resolutions / spring planning). Bikes are long-life purchases so demand concentrates early.

Recommendation (priority):

1. Plan inventory & promotions to build stock 1–2 months before Q1 peak.
2. Roll out Q1-focused marketing budget and pre-season campaigns starting late Q4.
3. Introduce off-season offerings: services, upgrades, accessories bundles, and servicing to smooth revenue.

4.0 CONCLUSION

Working on this project has been an enriching and rewarding experience. It allowed me to apply technical skills across SQL, Excel, data modeling, and dashboard design while strengthening my ability to transform raw data into meaningful business insights. Beyond the technical work, this project helped me deepen my understanding of end-to-end data analysis workflows and reinforced the value of structured, evidence-based decision-making. Overall, it has been a valuable learning journey, and I look forward to building even more advanced analytics solutions in the future.